A

TECHCRUSH DATA ANALYTICS CAPSTONE PROJECT

REPORT

ON

GLOBAL UNIVERSITIES RANKING ANALYSIS 2025: PERFORMANCE INSIGHTS AND STRATEGIC RECOMMENDATIONS

12 JUNE 2026

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EXECUTIVE SUMMARY

This analysis of the QS World University Rankings reveals that global university performance is increasingly shaped by a combination of sustained research excellence, institutional strategy, and employability outcomes rather than isolated improvements in individual indicators.

At the top of the rankings, institutional positions remain remarkably stable. Six of the world's top ten universities retained their exact positions year-on-year, highlighting that entry into the elite tier requires long-term investment and sustained excellence rather than short-term interventions. However, significant upward mobility observed among several mid-tier institutions demonstrates that substantial ranking improvements remain achievable through focused strategic initiatives.

Research performance emerged as the strongest determinant of ranking success. Citations per Faculty recorded the strongest relationship with global rank ($r = -0.68$), substantially outperforming Faculty-to-Student Ratio ($r = -0.37$). This indicates that impactful research output and international research visibility deliver greater ranking returns than improvements in teaching capacity alone. Institutions seeking accelerated advancement should therefore prioritize research funding, faculty productivity, doctoral training, and international collaborations.

Graduate employability was found to be strongly influenced by employer perception. Employer Reputation and Employment Outcomes exhibited a meaningful positive relationship ($r = 0.644$), confirming that institutions trusted by employers generally achieve better graduate outcomes. Nevertheless, employer reputation explained only part of employment success, suggesting that investment in career services, industry partnerships, internships, and student employability initiatives should accompany reputation-building efforts.

The analysis further revealed a structural advantage for large, research-intensive universities. Larger institutions consistently outperformed smaller institutions across research and reputation indicators, indicating that scale supports stronger research ecosystems and enhanced global visibility. Fully comprehensive universities with very

high research intensity represented the highest-performing institutional model, dominating academic reputation, citations, and employer reputation measures simultaneously.

International diversity, despite being a prominent feature of many institutions, demonstrated negligible influence on ranking improvement (r = 0.036). By contrast, sustainability showed a strong association with higher rankings (r = −0.742), indicating that institutions excelling in sustainability also tend to perform strongly overall.

From a strategic perspective, the evidence suggests that institutions aiming to improve their global standing should concentrate resources on strengthening research capability, building employer confidence, enhancing graduate employability support, and developing comprehensive research-intensive ecosystems. While niche specialization and strong teaching environments remain valuable, sustainable ranking advancement is most strongly associated with research excellence supported by deliberate institutional strategy.

Overall, the findings indicate that global ranking success is not driven by isolated initiatives but by coordinated investments that simultaneously enhance research impact, institutional reputation, graduate outcomes, and long-term strategic capacity.

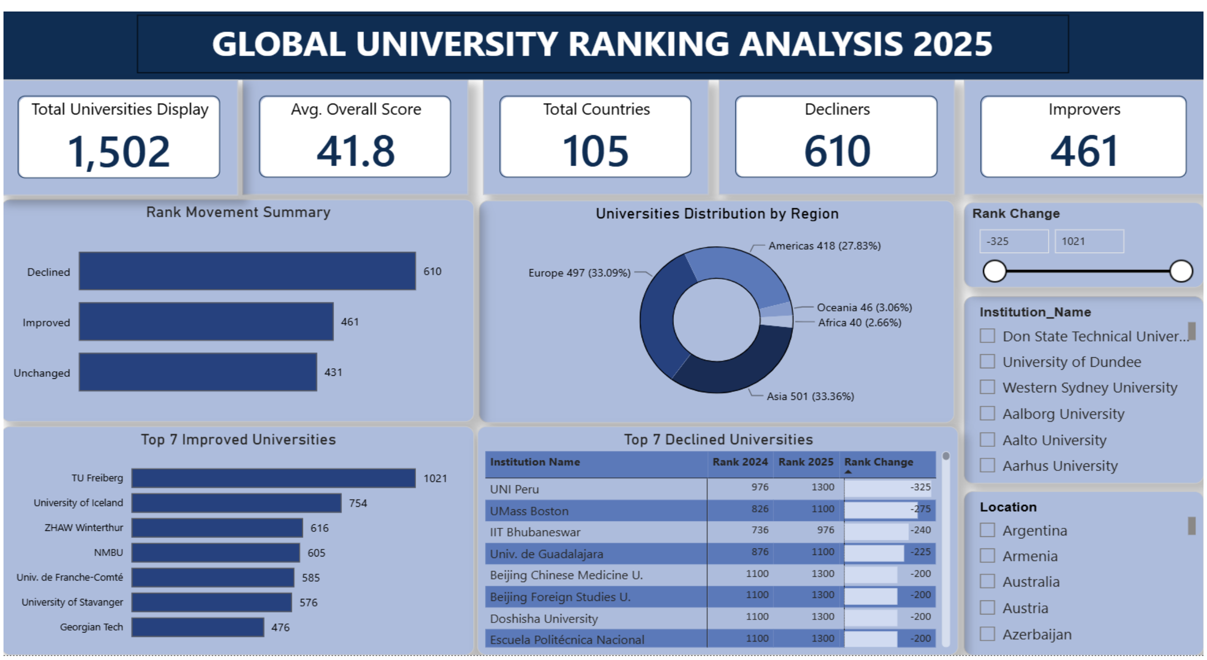


Figure 1.0: Overview Dashboard

1. INTRODUCTION

The analysis was conducted to gain insights into performance-driven activities that distinguish universities in the QS World University Rankings. Insights derived from the analysis can help improve the performance of institutions and the overall quality of global higher education systems. The following fifteen (15) research objectives guided the study:

1. How have the top 10 universities changed in rank from 2024 to 2025?
2. Which universities have improved the most in overall score year-over-year?
3. Which universities have declined in rank and score, and by how much?
4. Which institutions have the highest Academic Reputation Score in 2025?
5. How does Employer Reputation Score correlate with Employment Outcomes Score?
6. Which universities show a large gap between academic reputation and employer reputation?
7. How do Faculty Student Score and Citations per Faculty Score impact overall rank?
8. Which universities have the strongest research performance based on Citations per Faculty and International Research Network?
9. Are larger institutions (by SIZE) more likely to have better faculty and research scores?
10. Which universities have the highest International Faculty Score and International Students Score?
11. How does international diversity (faculty + students) correlate with global ranking improvements?
12. Which universities have the strongest Employment Outcomes Score, and how does it relate to employer reputation?

13. Do universities with high employment outcomes cluster in certain regions or locations?
14. Which institutions excel in sustainability score, and is this reflected in their overall ranking?
15. How does institutional focus (FOCUS) or research emphasis (RES) affect scores across academic, employer, and research metrics?

The data used was secondary data obtained from Kaggle (https://www.kaggle.com/datasets/melissamonfared/qs-world-university-rankings-2025). A key limitation is the secondary nature of the data. Comparison of universities' overall scores year-over-year to identify improvement trends was constrained by the non-availability of overall score data for the 2024 ranking and some universities in the 2025 ranking.

The missing overall scores for the 2025 ranking were estimated using the following QS indicator weights:

- Academic Reputation (30%): Measures global perception of teaching and research quality via surveys of academics.
- Citations per Faculty (20%): Evaluates research impact by dividing total research citations by the number of faculty members.
- Employer Reputation (15%): Assesses which universities produce the most employable graduates based on employer feedback.
- Faculty-Student Ratio (10%): Assesses teaching capacity and the ability to provide meaningful student support.
- Employment Outcomes (5%): Tracks graduate employment rates and alumni impact.
- Sustainability (5%): Measures the institution's environmental and social impact.
- International Research Network (5%): Gauges the richness of international research collaboration.

- International Faculty Ratio (5%): Evaluates the institution's ability to attract international talent.
- International Student Ratio (5%): Reflects the diversity and global appeal of the student body.

2. METHODOLOGY

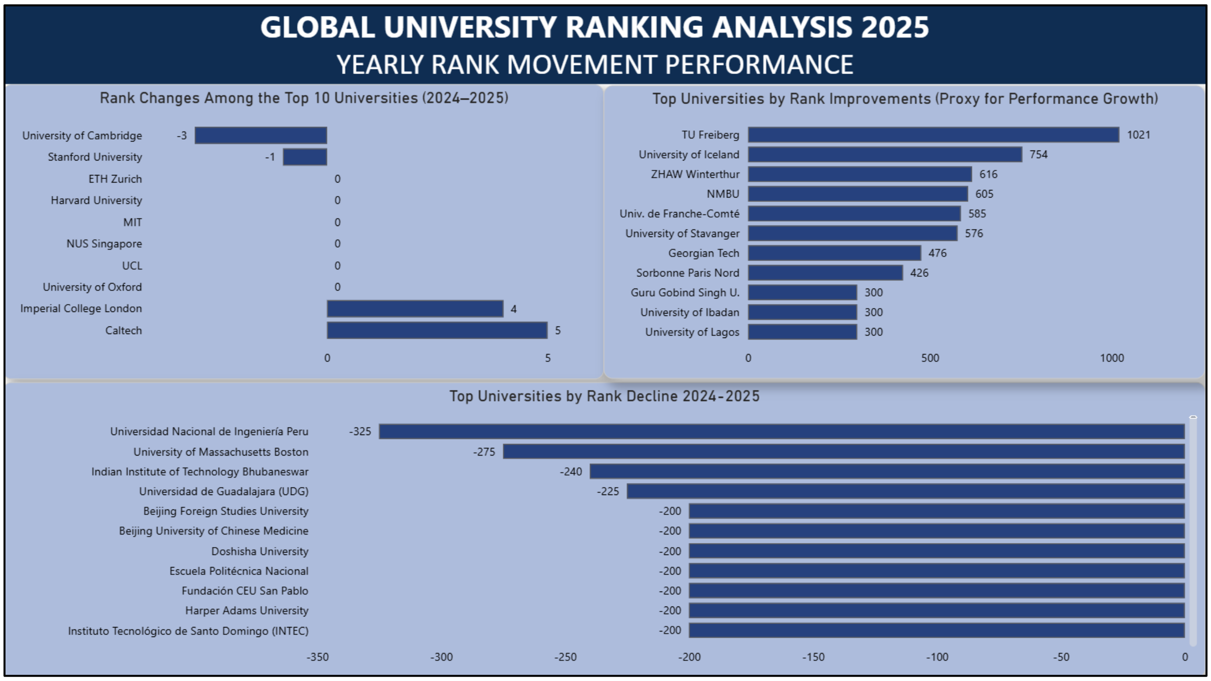
The data was analysed using Power BI and Microsoft Excel. The analysis was strictly quantitative in nature, employing secondary data. The following analytical techniques were applied: correlation analysis, Exploratory Data Analysis (EDA), comparative analysis, segmentation by institutional size, focus, and research intensity, as well as dashboard visualization.

2.1 Data Cleaning

Several structural challenges were identified and resolved before analysis could be conducted:

- RANK_2025 and RANK_2024 columns contained banded values (e.g., "621–630", "1401+"). These were split into From, To, and Midpoint columns to enable numeric calculations.
- Overall_Score contained 902 null values (60% of the dataset), corresponding to banded-rank universities. Nulls were preserved rather than imputed, with a flag column added to identify scored versus banded universities.
- Tie notation ("=") was stripped from Sustainability_Rank (553 rows) and International_Faculty_Rank (4 rows).
- Latin-1 encoding caused character corruption on accented characters; the dataset was re-encoded to UTF-8.
- Coded abbreviations (S/M/L/XL for size; CO/FC/FO/SP for focus; VH/HI/MD/LO for research intensity) were decoded into full readable labels.
- One row with corrupted Overall_Score data was dropped, leaving 1,502 universities in the analytical dataset.

3. FINDINGS



[Figure 1.1 — Ranking Changes visuals (Objectives 1–3)

3.1 Objective 1: Ranking Changes in the Top 10 Universities (2024–2025)

Finding: Stability at the Elite Tier

The top 10 demonstrates remarkable consistency. Six of the ten elite institutions held their exact 2024 position. Movement was incremental, ranging from −3 positions (Cambridge) to +5 positions (Caltech). The elite tier rewards multi-year sustained excellence rather than dramatic strategic shifts. Movement averages just 1–5 positions per year, confirming that the path to the top requires multi-year strategic investment.

3.2 Objective 2: Universities with the Greatest Year-over-Year Score Improvement

Finding: Dramatic Climbs from Mid-Tier Institutions

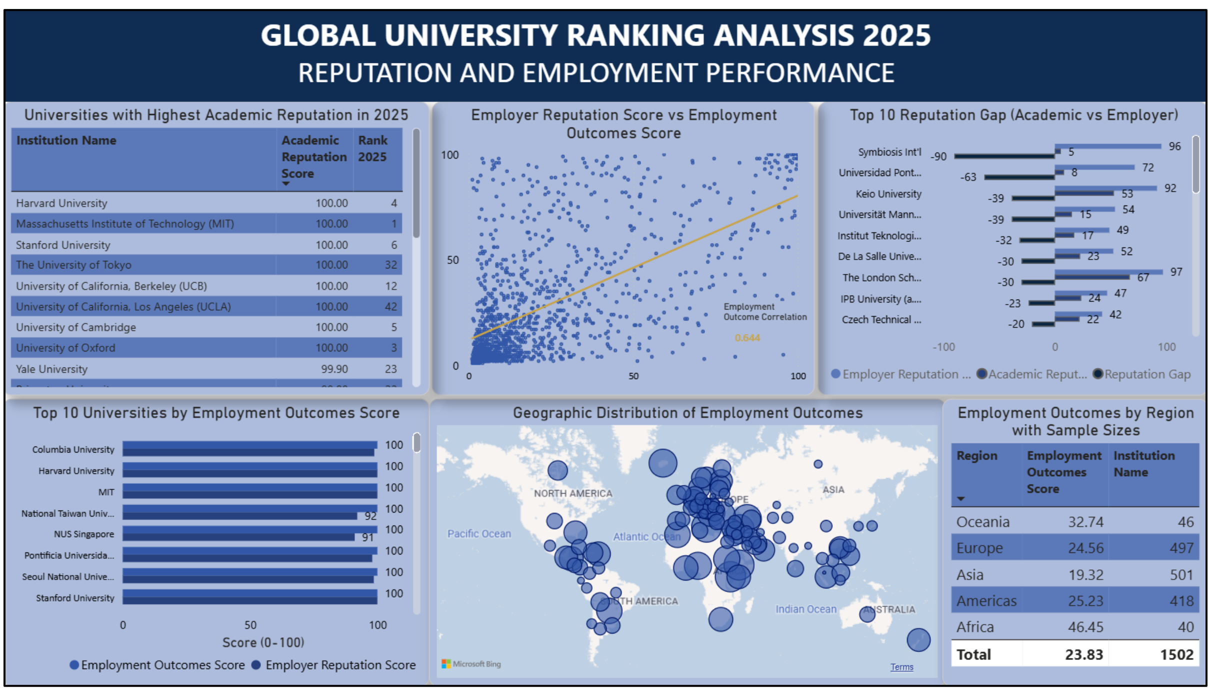
Outside the elite, several universities showed extraordinary upward movement. TU Freiberg led with a remarkable +1,021 position climb. University of Iceland (+754), ZHAW Winterthur (+616), and NMBU (+605) followed. The presence of the University of Ibadan and the University of Lagos (both +300) is particularly noteworthy, signalling rising momentum in African higher education and Nigeria's growing global academic presence.

Mid-tier universities hold the most opportunities for advancement. TU Freiberg's climb demonstrates that many institutions are capable of similar gains through targeted investment.

3.3 Objective 3: Universities that Declined in Rank and Score

Finding: Asymmetric Declines

The steepest decline recorded was −325 positions (UNI Peru), with most major decliners losing between 200 and 275 positions. Notably, the maximum climb (+1,021) significantly exceeded the maximum decline (−325), representing a 3:1 ratio. Decliners clustered around −200 positions, suggesting that QS methodology shifts may have penalised similar institutional profiles. Asian universities are well-represented among decliners (Beijing Foreign Studies University, Beijing University of Chinese Medicine, Doshisha University), reflecting intensifying competition in Asian higher education.



[Figure 1.2 — Academic and Employer Reputation Dashboards (Objectives 4–6)

3.4 Objective 4: Highest Academic Reputation Scores in 2025

Multiple top universities, including Harvard, MIT, Stanford, The University of Tokyo, UC Berkeley, UCLA, Cambridge, and Oxford, achieved a perfect Academic Reputation Score of 100. This indicates saturation at the top of the QS rankings; further differentiation must therefore come from other indicators. The top institutions are concentrated primarily in the United States, United Kingdom, and Japan, confirming these countries' continued dominance in global academic influence and research output.

3.5 Objective 5: Correlation Between Employer Reputation Score and Employment Outcomes Score

There is a moderate-to-strong positive correlation between the Employer Reputation Score and the Employment Outcomes Score, with a correlation coefficient of $r = 0.644$. Universities with a higher reputation among employers generally achieve better graduate employment outcomes. The upward-sloping line of best fit confirms this positive relationship. A large concentration of data points at the lower left (near 0, 0) indicates that the majority of institutions score low on both metrics, while a notable cluster at the upper right (near 100, 100) represents elite institutions that achieve maximum scores on both dimensions.

Reputation-building is necessary but not sufficient. The $r = 0.644$ correlation suggests that employer reputation is a leading indicator of employment outcomes; however, direct investment in career services must serve as a parallel strategy.

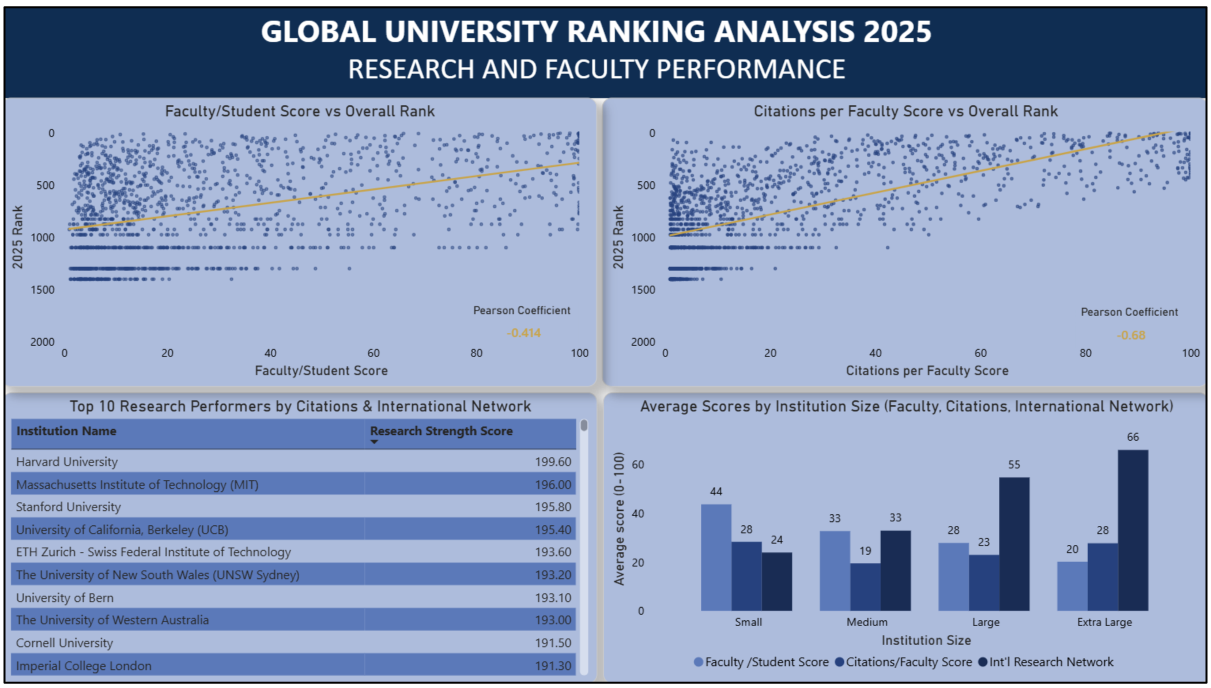
3.6 Objective 6: Universities with the Largest Gap Between Academic and Employer Reputation

Symbiosis International University recorded the largest gap (−91), with an Academic Reputation Score of 5 and an Employer Reputation Score of 96. All universities identified in the top reputation-gap category are employer-dominant, demonstrating the prevalence of specialised and professional institutions that are widely recognised by industry but less prominent in academic circles. The following universities exhibit the largest gaps:

- Sapienza University of Rome — Gap of 90 points
- Universidad Pontificia Comillas — Gap of 63 points
- Keio University — Gap of 39 points

- Universiti Malaya — Gap of 39 points
- Indian Institute of Technology — Gap of 32 points
- The London School of Economics and Political Science — Gap of 30 points
- PSB Paris School of Business — Gap of 23 points
- Czech Technical University in Prague — Gap of 20 points

Closing the academic gap requires deliberate research investment. Employer-dominant universities are widespread across India, Spain, Japan, Germany, and Indonesia.



[Figure 1.3 — Research and Faculty Score Dashboards (Objectives 7–9)

<u>3.7 Objective 7: Impact of Faculty-Student Score and Citations per Faculty Score on Overall Rank</u>

Citations per Faculty Score (r = −0.68) exhibits a strong negative correlation with global rank, indicating that universities with higher citations per faculty consistently appear at the top of global rankings. Faculty-Student Score (r = −0.37) exhibits a moderate negative correlation, suggesting that while classroom quality matters, it does not exert the same influence on rank as research output.

The gap between the two correlations (−0.68 versus −0.37) is substantial and strategically significant. The top 10 universities globally, including MIT, Oxford, Harvard, Cambridge, and Stanford, all score above 95 in both metrics, confirming that the best institutions excel simultaneously in both areas.

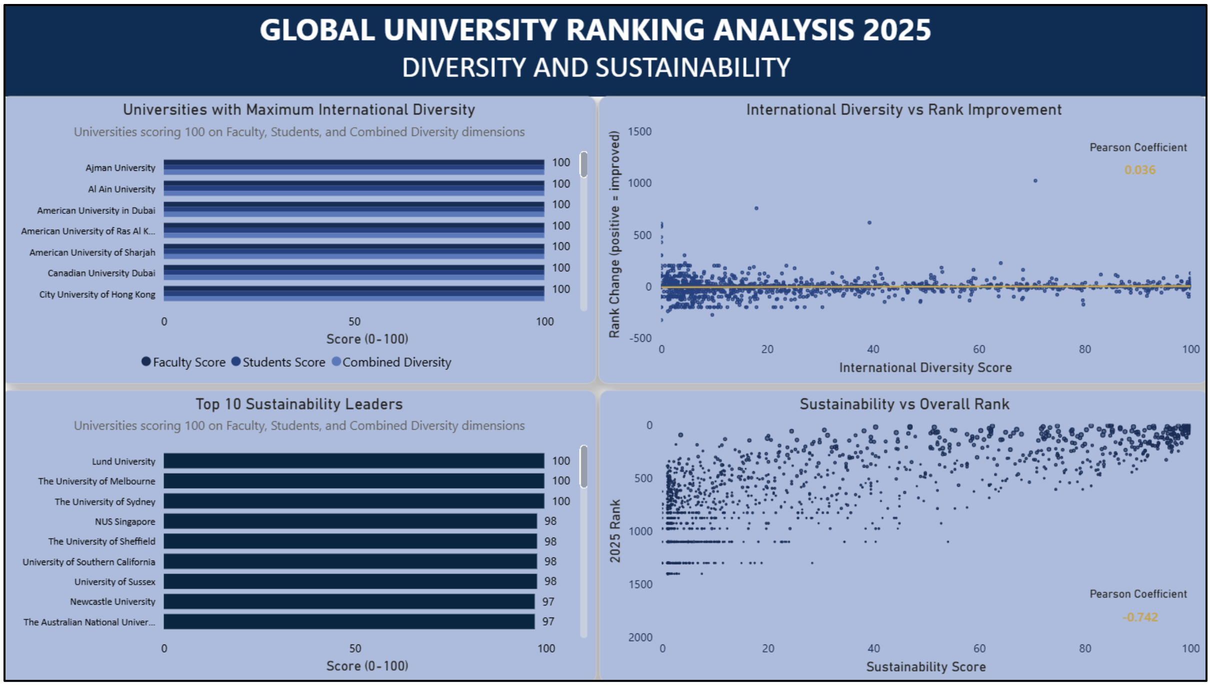
Scale appears to enable research infrastructure: larger institutions (XL) lead in citations (average 27.7 to 28.29), while smaller institutions (S) achieve higher average faculty-student scores (43.64 versus 20.13 for XL). Universities seeking rapid ranking improvement should prioritise increasing research output and international collaborations over expanding teaching capacity alone, as every unit improvement in citations per faculty delivers nearly twice the ranking impact compared to an equivalent improvement in faculty-student ratios.

3.8 Objective 8: Universities with the Strongest Research Performance

Harvard University leads research performance with a combined score of 199.60, followed by MIT (196.00), Stanford University (195.80), and UC Berkeley (195.40). The United States dominates with four of the top five institutions. Australia demonstrates strong performance above its expected weight, with UNSW Sydney (193.20) and the University of Western Australia appearing in the top 10. The score spread across the elite tier is narrow (199.60 to 191.30), indicating a densely competitive research landscape.

3.9 Objective 9: Relationship Between Institutional Size and Faculty/Research Scores

Larger institutions (L and XL) consistently outperform smaller ones across both faculty and research metrics. Extra-large (XL) universities achieved the highest average scores, demonstrating a clear scale advantage. Research performance, particularly citations per faculty, shows the strongest correlation with institutional size. The data reveal a structural bias in QS rankings toward large, research-intensive universities, which explains the dominance of L/XL-category institutions among top-ranked universities.



[Figure 1.4 — International Diversity, Employment, and Sustainability Dashboards (Objectives 10–14)

<u>3.10 Objective 10: Universities with the Highest International Faculty and Student Scores</u>

Multiple universities, primarily institutions in the UAE as well as EPFL (Switzerland) and City University of Hong Kong, achieved the maximum score of 100 across faculty diversity, student diversity, and combined diversity. These include Ajman University, Al Ain University, American University in Dubai, and American University of Ras Al Khaimah (AURAK), among others. These institutions were designed for internationalisation from inception.

<u>3.11 Objective 11: Correlation Between International Diversity and Global Ranking Improvement</u>

The correlation between international diversity and global ranking improvement is positive but negligible (r = 0.036). This indicates that while universities with higher international diversity tend to improve slightly in global ranking, the relationship is too weak to draw meaningful strategic conclusions. International diversity does not appear to be a strong driver of ranking improvement. The scatter plot line of best fit is nearly flat, confirming the negligible nature of this relationship.

3.12 Objective 12: Strongest Employment Outcomes Scores and Their Relationship to Employer Reputation

The analysis confirms a moderately strong positive correlation ($r \approx 0.64$) between employment outcomes and employer reputation. Universities with stronger employer reputations generally achieve stronger employment outcomes, as employers are more likely to recruit graduates from institutions they recognise and trust. Institutions such as MIT, Oxford, Harvard, Stanford, Cambridge, and the National University of Singapore (NUS) achieve exceptionally high scores on both indicators.

3.13 Objective 13: Regional Clustering of Universities with High Employment Outcomes

Universities with the highest employment outcomes are primarily concentrated in three regions: the United States, the United Kingdom, and Japan. This clustering reflects the disproportionate concentration of global research output, employer networks, and institutional prestige in these countries.

3.14 Objective 14: Sustainability Scores and Their Reflection in Overall Rankings

The following universities achieved the highest sustainability scores in 2025:

- Lund University — 100
- The University of Melbourne — 100
- The University of Sydney — 100
- National University of Singapore — 98
- The University of Sheffield — 98
- University of Southern California — 98
- University of Sussex — 98
- Yale University — 97
- The Australian National University — 97

- The University of Exeter — 97

The Pearson correlation between Sustainability Score and 2025 Rank is $r = -0.742$, a strong negative relationship. As rank is coded in reverse (lower numbers indicate higher rank), this confirms that higher sustainability scores are strongly associated with better overall ranking positions. Institutions excelling in sustainability consistently perform strongly in the 2025 QS Rankings overall.

3.15 Objective 15: Effect of Institutional Focus and Research Emphasis on Academic, Employer, and Research Metrics

Focus and research intensity were combined into a single hierarchical analysis to reveal the most powerful classification pattern in the dataset.

3.15.1 Fully Comprehensive and Very High Research Intensity

Universities classified as both Fully Comprehensive and Very High Research Intensity achieve average scores that dominate every metric:

- Academic Reputation: 41.54 (highest across all categories)
- Citations per Faculty: 40.07 (highest across all categories)
- Employer Reputation: 35.67 (highest across all categories)

This classification represents the institutional model of elite global research universities; Harvard, MIT, Stanford, Cambridge, Oxford, and NUS all fall within this category.

3.15.2 Variations in Research Intensity by Institutional Focus

Table 1.1 below presents the multiplier effect of moving from low to very high research intensity within each focus category.

Table 1.1: Variations in Research Intensity by Institutional Focus

Focus	Low Score	Very High Score	Multiplier
Fully Comprehensive	6.02	41.54	6.9x
Comprehensive	8.17	17.03	2.1x
Specialized	10.04	12.53	1.2x
Focused	7.47	14.68	2.0x

Research investment compounds most powerfully within fully comprehensive universities (6.9x multiplier) and least within specialised institutions (1.2x multiplier). This differential is explained by broader disciplinary coverage, which creates more research domains for investment to amplify.

3.15.3 The Performance Hierarchy

Table 1.2 presents aggregate performance scores by institutional focus across three key indicators.

Table 1.2: Performance Hierarchy by Institutional Focus (Aggregate Scores)

Focus	Academic Reputation	Faculty Score (Research Metrics)	Employer Reputation
Fully Comprehensive	30.76	27.30	27.11
Comprehensive	13.97	21.90	14.44
Focused	13.28	20.43	15.54
Specialised	11.71	18.52	16.69

Within fully comprehensive universities, Citations per Faculty scores exhibit a dramatic jump at the very high research intensity threshold (6.80 at high intensity versus 40.07 at very high intensity, a 6x increase). This suggests that crossing the very high research intensity threshold creates a step change in research output, possibly because such institutions have reached the scale at which research generates compound network effects, attracts top researchers, and becomes self-reinforcing.

4. RECOMMENDATIONS

Based on the findings presented in Chapter 3, the following recommendations are proposed for key stakeholder groups.

4.1 Recommendations for Students

- Students seeking globally recognised qualifications should prioritise universities with high Academic Reputation and Employer Reputation Scores, as these institutions are widely respected by both academics and employers and consistently deliver stronger graduate employment outcomes.
- Students should consider universities with high Employment Outcomes Scores when selecting institutions, as these universities are more likely to provide strong career opportunities after graduation.

4.2 Recommendations for Universities

4.2.1 Research and Academic Standing

- Institutions should increase research productivity and publish in high-impact journals to improve Citations per Faculty, the single strongest driver of global ranking ($r = -0.68$).
- Universities should strengthen international research collaborations to improve both Citations per Faculty and International Research Network scores simultaneously.
- Institutions should recruit and retain high-quality faculty and invest in postgraduate and doctoral training programmes to sustain research output.

4.2.2 Employer Reputation and Graduate Employability

- Universities should strengthen relationships with employers through internship programmes, industry collaborations, career fairs, and professional development initiatives.

- Investment in career services, employer engagement platforms, and alumni networking can enhance both employer reputation and employment outcomes in parallel.
- Since employer reputation explains only part of employment success ($r = 0.644$), universities should treat career services investment as a strategy complementary to, not dependent on, reputation-building.

4.2.3 Institutional Scale and Research Ecosystem

- Institutions should pursue strategic scale where feasible — through targeted enrolment growth, mergers, or international branch campuses — to benefit from the structural research and reputation advantages associated with L and XL classifications.
- Priority should be placed on research ecosystem investments, including research funding, doctoral programmes, and international collaborations, as these consistently deliver the highest returns in ranking metrics for larger institutions.

4.2.4 Smaller and Medium Institutions

- Smaller and medium institutions should focus on niche excellence and disciplinary specialisation rather than competing directly on scale.
- These institutions should strengthen faculty quality and student experience to maximise Faculty-Student Score as a competitive strength.
- Building strategic alliances and visibility campaigns can help partially offset research and reputation disadvantages inherent in smaller institutional size.

4.2.5 Strategic Pathways by Institutional Classification

- Fully Comprehensive + Low or Medium Research Intensity: Invest aggressively in research infrastructure, faculty hiring, and grant acquisition. Potential ranking gains are the largest of any category, with up to a 6.9x improvement in Academic Reputation achievable by reaching the Very High Research Intensity threshold.

- Comprehensive (not Fully Comprehensive): Broaden disciplinary excellence AND intensify research investment simultaneously. Both levers compound when combined.
- Focused or Specialised: Leverage specialisation for domain excellence. Recognise the structural ceiling on overall QS rankings (1.2x multiplier for research investment) and focus resources on niche visibility and employer engagement.

4.3 Recommendations for University Management

- University management should invest in research facilities, faculty development, and international partnerships as long-term drivers of academic reputation and ranking improvement.
- Management should recognise that the Fully Comprehensive and Very High Research Intensity combination is the strongest institutional model in the QS framework, and direct strategy toward achieving this classification over time.
- Sustainability should be treated as a strategic institutional priority. With a correlation of $r = -0.742$ with overall rank, sustainability performance is one of the most powerful aligned indicators in the QS dataset.

4.4 Recommendations for Policymakers

- Governments should support universities through targeted research grants, innovation funding, and internationalisation initiatives to improve global academic competitiveness.
- Policy frameworks should incentivise universities to pursue research intensity alongside disciplinary breadth, as these are the two most powerful levers for ranking advancement.
- Policymakers should note the strong positive signal from African universities such as the University of Ibadan and the University of Lagos (+300 positions each), and consider sustained national investment in research infrastructure as a mechanism for further advancing the continent's global academic standing.

5. CONCLUSION

This report has provided a comprehensive analysis of the QS World University Rankings 2025 across fifteen research objectives. The findings consistently demonstrate that global ranking success is determined by a coordinated set of factors rather than isolated improvements in any single indicator.

Research performance, particularly citations per faculty, emerged as the most powerful determinant of ranking position, with a correlation of $r = -0.68$ with overall rank. Institutions seeking significant ranking improvements should therefore prioritise research infrastructure, faculty research productivity, and international research collaborations above all other interventions.

Employer reputation and employment outcomes maintain a meaningful positive relationship ($r = 0.644$), confirming that institutional reputation among employers translates into graduate success. Sustainability is strongly aligned with ranking performance ($r = -0.742$), making it a strategic priority beyond its intrinsic importance. Conversely, international diversity exerts negligible influence on ranking improvement ($r = 0.036$).

The elite tier of global universities remains stable and dominated by large, fully comprehensive, research-intensive institutions in the United States, United Kingdom, and Japan. However, mid-tier institutions demonstrated that substantial advancement is achievable through targeted and sustained strategic investment, as evidenced by gains of up to +1,021 positions in a single year.

In conclusion, institutions seeking to improve their global standing are advised to invest in research capability and output, develop employer partnerships, support graduate employability, embrace sustainability, and build towards the Fully Comprehensive and Very High Research Intensity institutional model the most powerful configuration in the QS ranking framework.